

4_popquiz electric field 2

 This is a preview of the published version of the quiz

Started: Mar 15 at 3:09pm

Quiz Instructions

Note:

A vector E has a magnitude. The notation here will be $|E|$

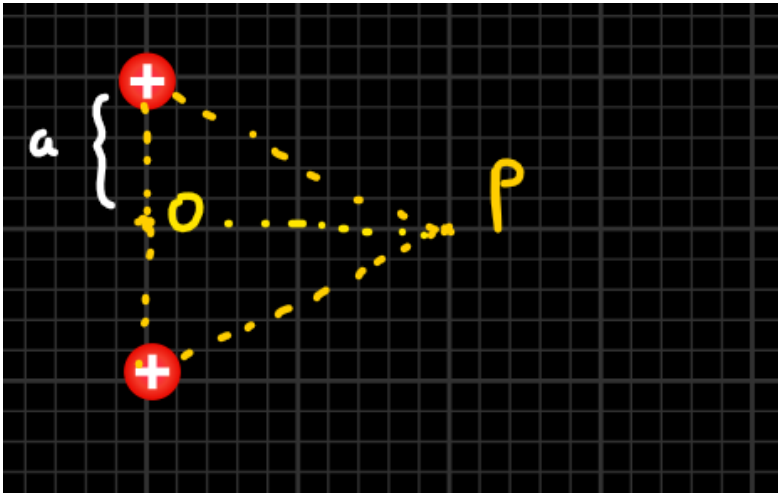
E is a vector and has components. Its represented by an arrow. The magnitude is the length of the arrow. And is found using Pythagorean theorem



Question 1 20 pts

Consider the following charge distribution (q_1 and q_2 have the same magnitude and are both positive)

At point P the net electric field E is



you can use the app if you want. Use the "sensor" or test charge. It will give you the direction of the electric field.

<https://phet.colorado.edu/en/simulations/charges-and-fields> 

(<https://phet.colorado.edu/en/simulations/charges-and-fields>)

E is the vector sum of E_1 and E_2

☐

is along the positive y-axis

☐

along the positive x-axis



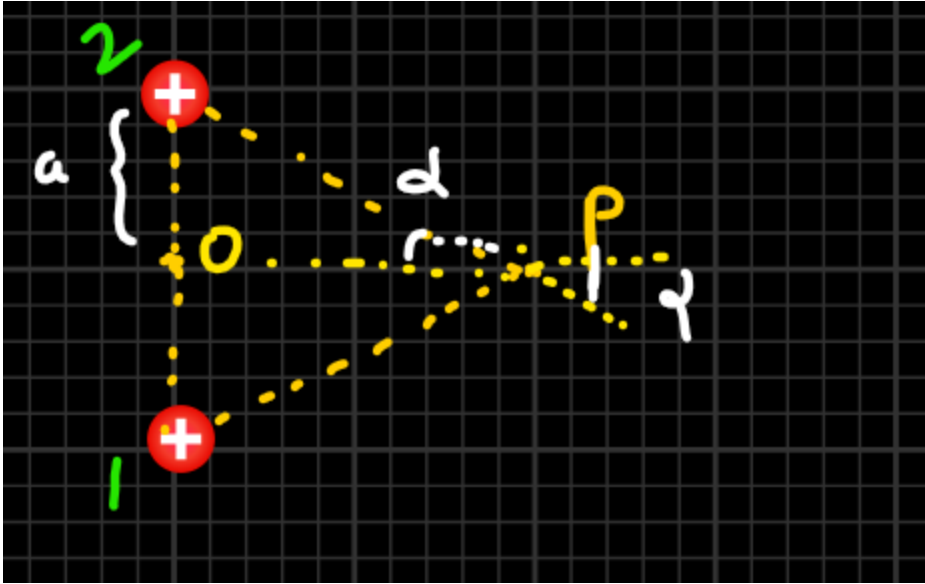
is in the first quadrant ($E_x > 0$, $E_y > 0$)



along the negative x-axis



Question 2 20 pts



The distance OP is called x . The distance between the 2 charges is $2a$.

Draw the electric field E_1 and E_2 generated by each of the charge. E_1 and E_2 are vectors. forces per unit charge applied by q_1 and q_2 .

E is the vector sum of $E_1 + E_2$. Draw E . (parallelogram method). E is a vector.

what is true (check with your classmates). $|E|$ is the magnitude of the electric field at P (length of the arrow)



$|E| = 2 |E_1| \cos(\alpha)$



$|E| = |E_1| = |E_2|$



$|E| = |E_1| \tan(\alpha)$



$E = 2 |E_1| \sin(\alpha)$



Question 3 20 pts

A simple and common technique for accelerating electrons is shown in Figure 18.55, where there is a uniform electric field between two plates. Electrons are released, usually from a hot filament, near the negative plate, and there is a small hole in the positive plate that allows the electrons to continue moving. (a) Calculate the acceleration of the electron if the field strength is $2.50 \times 10^4 \text{ N/C}$. (b) Explain why the electron will not be pulled back to the positive plate once it moves through the hole.

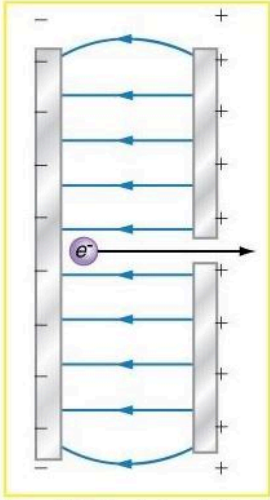


Figure 18.55 Parallel conducting plates with opposite charges on them create a relatively uniform electric field used to accelerate electrons to the right. Those that go through the hole can be used to make a TV or computer screen glow or to produce X-rays.

Find the acceleration of the electron.

Mass = $9.11 \times 10^{-31} \text{ kg}$. charge electron = $1.6 \times 10^{-19} \text{ C}$

Remember Newton's second law. $F = ma$ and here we have $|F| = q|E|$

☐

$1.7 \times 10^{11} \text{ m/s}^2$

☐

There is no acceleration

☐

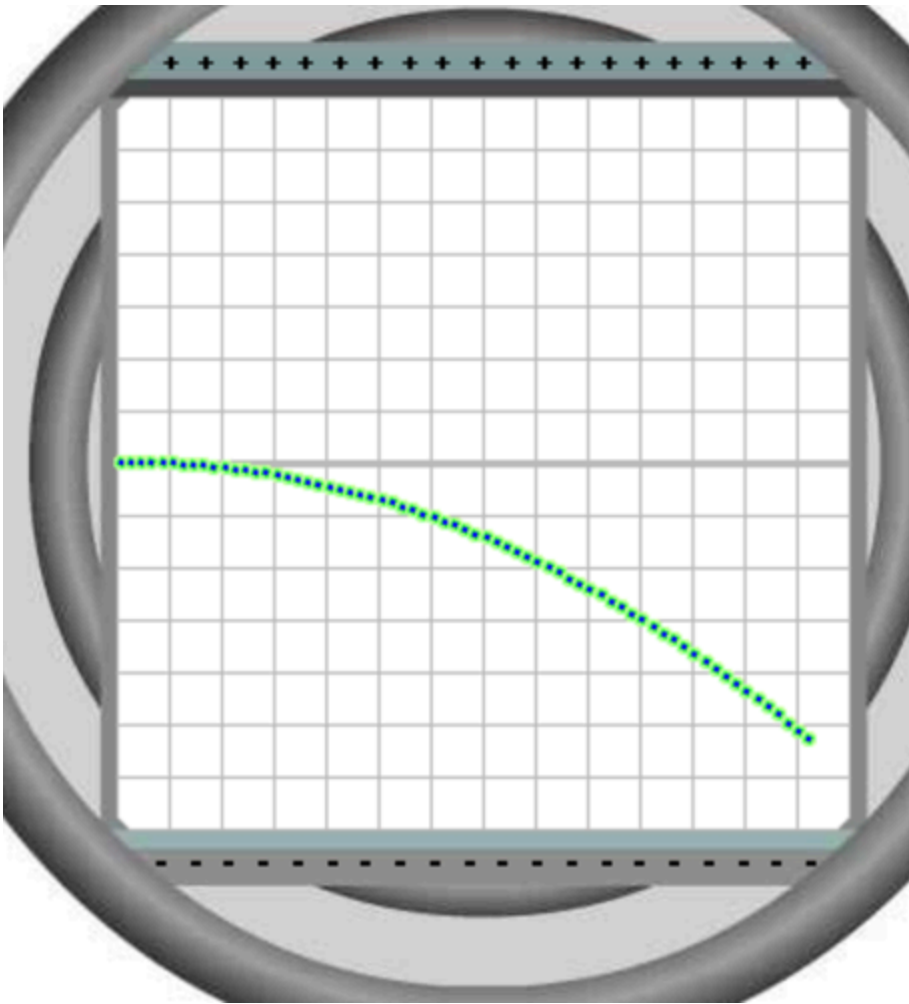
$4.4 \times 10^{15} \text{ m/s}^2$

☐

$2.7 \times 10^{34} \text{ m/s}^2$

☐

Question 4 20 pts



A charge is accelerated in a CRT (cathode ray tube) as shown. There is an electric field between the positively charged plate and a negatively charged plate. The initial velocity of the particle is along the x-axis.

What is true

☐

the charge is positive

☐

the charge is negative

☐

the particle is neutral

☐

can not be determined



Question 5 10 pts

Build on the previous questions. +xaxis is @ right and +yaxis is @ up

What is true about the vector acceleration $\mathbf{a}$ and its components a_x , a_y

hint: remember physics 1. acceleration is a vector. it is force per unit mass. $\mathbf{a} = \mathbf{F} / m$

so the acceleration has the same direction as the force. its magnitude is the magnitude of the force divided by the mass. but they have the same direction

☐

$a_x = q|E|/m$, $a_y = -q|E|/m$

☐

$a_x = 0$, $a_y = -|E|/m$

☐

$a_x = q|E|/m$, $a_y = 0$

☐

$a_x = 0$, $a_y = -q|E|/m$



Question 6 15 pts

The electric field

☐

diverges from a positive charge and converges in a negative charge

☐

converges to any charge. positive or negative

☐

electric charges don't generate electric fields

☐

converges in a positive charge and diverges from a negative charge

Not saved

Submit Quiz